

PASS CHEMISTRY BOOK 1510/1520



Lindsay Blackstock & Sharon Brewer
Thompson Rivers University

Thompson Rivers University
PASS Chemistry Book 1510/1520

Lindsay Blackstock & Sharon Brewer

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

This text was compiled on 06/08/2025

TABLE OF CONTENTS

[Licensing](#)

1: Introduction

- [1.1: About PASS and the Creators](#)
- [1.2: How to use PASS Chem](#)

2: Gases

- [2.1: PASS Ideal Gases- General gas law calculation, changing temperature \(2.E.12\)](#)
- [2.2: PASS Ideal Gases - Calculate the density of an ideal gas \(2.E.28\)](#)
- [2.3: PASS Ideal Gases - partial pressure, gas over water calculation \(2.E.36\)](#)
- [2.4: PASS Ideal Gases- use the Kinetic Molecular Theory to explain impact on a gas as V and T increased \(2.E.45\)](#)

3: Thermochemistry

- [3.1: PASS Thermochemistry: calculate q for cooling a compound \(3.E.14\)](#)
- [3.2: PASS Thermochemistry- Interpreting thermochemical equations and sketching enthalpy diagrams \(3.E.10a\)](#)

4: Kinetics

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Introduction

[1.1: About PASS and the Creators](#)

[1.2: How to use PASS Chem](#)

[1: Introduction](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

1.1: About PASS and the Creators

The PASS (Platform Adapted Strategic Solutions) Framework was developed to provide a literature-based strategy for OER practice problem solutions for general chemistry practice questions. The PASS solutions presented in these collections contain templated solutions developed for OER chemistry problems across first year chemistry topics at TRU that were adapted for the LibreText platform.



The content creators pictured above are Assistant Teaching Professor Dr. Lindsay Blackstock, Student Research Partner Ashlynn Jensen, and Associate Professor Dr. Sharon Brewer. The content development was made possible by funding from a [TRU Instructional Innovation Grant \(TIIG\)](#) awarded to Dr. Blackstock & Dr. Brewer in 2023. The PASS framework development was funded by a TRU Open Educational Resources Grant awarded to Dr. Blackstock, Dr. Brewer and [Dr. Verena Roberts](#) in 2021.

[1.1: About PASS and the Creators](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

1.2: How to use PASS Chem

This book follows the Platform Adaptable Strategic Solution (PASS) template designed by Drs. Lindsay Blackstock, Sharon Brewer, and Verena Roberts to support students at various levels of understanding and mimic the experience that a student may have during an office hour. You can find the first examples using this template on the PASS Chem website: <https://chempass.opened.ca/>.

For each question, students may attempt and check their answer with **Show/Hide Answer**. If you were correct, **AMAZING!** You can move on to the next question.

Strategy Map: If you need assistance in *what general steps to take* (i.e., spoiler-free!) then move on to the **strategy map**. The strategy map will provide you with a procedure (instructions) of what to do to reach the answer.

Solution: If you would like to see each step (i.e., spoiler alert!), move on to the **solution**. This section shows a simple step-by-step solution to reach the answer but without any commentary.

Guided Solution: If you need more help and want to know *why* you are performing each step in the solution, move on to the **guided solution**.

The guided solution explains the chemistry theory or rationale behind the process. The guided solution's purpose is to replicate what a student may experience during an office hour or a lecture, working through a problem supported by the 'teacher talk.' Here, you will also find supplemental information including hints embedded throughout, such as:

- Links to read more from the LibreTexts reference textbooks
- Additional context or an alternative perspective
- Reminders about various unit conversions or significant figures
- Common mistakes to avoid

Complete Solution: Below the guided solution is the **complete solution** that combines the step-by-step solution with the key context points from the guided solution.

Check Your Work: In chemistry, you can often avoid mistakes by thinking critically about the theory related to the problem and predicting what the answer *should or shouldn't be*. Be sure to **check your work** (see the purple box below). This section provides the rationale of why your answer 'makes sense' with respect to the chemistry theory. Use this strategy to avoid making mistakes in similar problems.

Our suggested approach:

1. Attempt it yourself. Check to see if you need additional assistance to get the correct answer.
2. After checking your answer, decide how much additional support you need to understand how to solve the problem yourself.

Navigate to one of the three boxes below:

1. **Strategy Map**
2. **Solution**
3. **Guided Solution.**

The boxes are listed in order of how much assistance is provided from least to most.

Below the answer is the link to the online resource to find supplementary information about this topic.

We hope this helps you build your confidence in working through these practice chemistry problems!

1.2: How to use PASS Chem is shared under a [CC BY-NC-SA 4.0](https://creativecommons.org/licenses/by-nc-sa/4.0/) license and was authored, remixed, and/or curated by LibreTexts.

CHAPTER OVERVIEW

2: Gases

2.1: PASS Ideal Gases- General gas law calculation, changing temperature (2.E.12)

2.2: PASS Ideal Gases - Calculate the density of an ideal gas (2.E.28)

2.3: PASS Ideal Gases - partial pressure, gas over water calculation (2.E.36)

2.4: PASS Ideal Gases- use the Kinetic Molecular Theory to explain impact on a gas as V and T increased (2.E.45)

2: Gases is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

2.1: PASS Ideal Gases- General gas law calculation, changing temperature (2.E.12)

This work by Jensen, Brewer, Blackstock is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



? Exercise 2.E.12

A spray can which has a pressure of 1344 torr at 23°C is used until it is empty except for the propellant gas. If the can is thrown into a fire at 475°C, what will be the pressure in the hot can in atmospheres?

Answer

Final pressure = 4.47 atm

Strategy Map

Step	Hint
1. Identify the information given in the question and what conditions are changing.	
2. Identify what the question is asking you to calculate and what variable it would be represented by.	
3. Choose an equation that compares the correct relationships.	You will need to manipulate the equation to compare the initial and final conditions.
4. Do any necessary conversions to ensure you are using the appropriate units.	Calculations for ideal gases using temperature must have the temperature in Kelvin. Since we want our final pressure in atmospheres we can convert the initial pressure to atmospheres in our calculation.

Solution

$$\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$$

$$\frac{P_i T_f}{T_i} = P_f$$

Torr → atmospheres

$$\frac{1344 \text{ torr}}{1} \times \frac{1 \text{ atm}}{760 \text{ torr}} = \mathbf{1.768 \text{ atm}}$$

Degrees Celsius → Kelvin

$$23^\circ\text{C} + 273.15 = \mathbf{296\text{K}}$$

$$475^\circ\text{C} + 273.15 = \mathbf{748\text{K}}$$

$$\frac{P_i T_f}{T_i} = P_f$$

$$\frac{(1.77 \text{ atm})(748\text{K})}{296\text{K}} = P_f$$

$$\mathbf{P_f = 4.47 \text{ atm}}$$

Guided Solution

[Download Guided Solution as a pdf](#)

Guided Solution

Hint

	Guided Solution	Hint
tion	This is a calculation type problem where we use the ideal gas laws to calculate how the pressure would change due to a change in conditions.	Refer to: LibreText Section 2.3
tion	A spray can which has a pressure of 1344 torr at 23°C is used until it is empty except for the propellant gas. If the can is thrown into a fire at 475°C, what will be the pressure in the hot can in atmospheres?	We are told that the initial pressure is 1344 torr and that the initial temperature is 23°C. We are also given the final temperature which is 475°C. We also know that the amount of gas inside the can stays constant before and after we are evaluating its pressure.
tion	The question asks you to find the pressure after the can is heated.	
tion	If the can is being heated you can expect the final pressure to be higher than the initial pressure.	If your calculated final pressure is lower than the initial pressure, you must have made an error somewhere.
tion	Recall the ideal gas equation. $PV = nRT$ How can we manipulate it to find the information we want?	We know that the ideal gas equation is $PV = nRT$ where 'R' is a constant. Because this value is constant, we can isolate it and create a derived equation where the initial values equal the final values. $\frac{P_i V_i}{n_i T_i} = R = \frac{P_f V_f}{n_f T_f}$ $\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$ Now we can get rid of values that stay constant during this problem. In this case the Volume and number of moles stays constant so it can be removed. (it can be divided out since it is the same on both sides). $\frac{P_i T_f}{T_i} = P_f$ Once we isolate for our desired variable, we get our new equation which we can plug in and use to solve for the final pressure.
tion	Before plugging into your equation ensure you are using values with the correct units. If necessary, use conversion factors.	Since we want our final pressure in atmospheres we will convert the initial pressure to atmospheres. Since the pressure of an ideal gas is proportional to temperature in Kelvin, we must convert our temperatures to Kelvin in order to use our derived relationship.

	Guided Solution	Hint
tion	Complete Solution Conversions: Torr → atmospheres $\frac{1344 \text{ torr}}{1} \times \frac{1 \text{ atm}}{760 \text{ torr}} = \mathbf{1.768}$ Degrees Celsius → Kelvin $23^{\circ}\text{C} + 273.15 = \mathbf{296k}$ $475^{\circ}\text{C} + 273.15 = \mathbf{748k}$ Manipulating the Ideal gas equation: $PV = nRT$ R is a constant, we can rearrange and isolate 'R.' Since it is a constant, the initial conditions and final conditions will both equal the 'R.' $\frac{P_i V_i}{n_i T_i} = R = \frac{P_f V_f}{n_f T_f}$ Since they both equal 'R' they also equal each other. $\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$ Now we can cancel out the variables with values that do not change for this problem. In this case the volume and number of moles do not change. We can simply remove them from our equation. $\frac{P_i}{T_i} = \frac{P_f}{T_f}$ Now all that's left is to isolate the desired variable, P_f. $\frac{P_i T_f}{T_i} = P_f$ Finally, plug your initial and final temperatures and initial pressure into the derived equation, and solve for the final pressure: $\frac{P_i T_f}{T_i} = P_f$ $\frac{(1.77 \text{ atm})(748k)}{296k} = P_f$ $P_f = 4.47 \text{ atm}$	
tion	Check your work! We expected the final pressure to be higher than the initial pressure, and it is: $P_i = 1.77 \text{ atm}$, $P_f = 4.47 \text{ atm}$ Why does this answer make chemical sense? We know from Amonton's Law that temperature and pressure are proportional, meaning as temperature increases, the pressure will also increase. Using this knowledge, we know that our calculation must show that the final pressure is higher than the initial pressure.	If our calculation shows the final pressure as unchanged or less than the initial pressure, an error likely occurred in the calculation process.

(question source from page titled 9.E:Gases (Exercises)
[https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_\(OpenSTAX\)/09%3AGases/9.E%3AGases_\(Exercises\)](https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_(OpenSTAX)/09%3AGases/9.E%3AGases_(Exercises)),
 shared under a CC BY 4.0 license, authored, remixed, and/or curated by OpenStax, original source
<https://openstax.org/books/chemistry/pages/9-exercises>, Access for free at <https://openstax.org/books/chemistry/pages/1-introduction>)

2.1: PASS Ideal Gases- General gas law calculation, changing temperature (2.E.12) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

2.2: PASS Ideal Gases - Calculate the density of an ideal gas (2.E.28)

This work by Jensen, Brewer, Blackstock is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



? Exercise 2.2.2.E.28

What is the density of laughing gas, dinitrogen monoxide, N_2O , at a temperature of 325 K and a pressure of 113.0 kPa?

Answer

1.84g/L

Strategy Map

	Step	Hint
	1. Identify what information the question gives you and do any necessary conversions.	See LibreText 2.4, Stoichiometry of Gaseous Substances, Mixtures, and Reactions
if ...	2. There are two approaches you could use: Approach a. To do this in one-step you can derive an equation that contains all known variables and the unknown density by manipulating the ideal gas equation to obtain an expression for density.	- Recall the density equation. - Recall n is amount in moles which is equal to mass/molar mass. - Substitute n for $\frac{m}{M}$ in the ideal gas equation. - Manipulate the ideal gas equation (move the variables around) so that you can solve for density.
if ...	Approach b. To do this in two steps first use the ideal gas equation to solve for volume. Then use the density equation to solve for density using the calculated volume.	$V = \frac{nRT}{P}$ $d = \frac{m}{V}$
if ...	3. Plug data in to equation and solve for your density.	Make sure to write your units in for all values.

Solution

$$113.0\text{kPa} = \frac{1\text{ atm}}{101.325\text{kPa}} = 1.115\text{ atm}$$

Approach a, using a derived equation:

$$d = \frac{\text{mass}}{\text{volume}} = \frac{m}{V}$$

$$PV = nRT$$

$$n = \frac{\text{mass}}{\text{molar mass}} = \frac{m}{M}$$

$$PV = \frac{m}{M}RT$$

$$d = \frac{m}{V} = \frac{PM}{RT}$$

$$d = \frac{(1.115\text{ atm})(44.013\text{ g/mol})}{(0.08206\text{ atm} \cdot \text{L/mol} \cdot \text{k})(325\text{ k})}$$

$$d = 1.84\text{ g/L}$$

Approach b, using a two steps:

Step one

$$PV = nRT$$

$$V = \frac{nRT}{P}$$

$$V = \frac{(1.000 \text{ mol})(0.08206 \text{ atm} \cdot \text{L/mol} \cdot \text{k})(325\text{k})}{1.115 \text{ atm}}$$

$$V = 23.9 \text{ L}$$

Step two

$$d = \frac{m}{V}$$

$$d = \frac{44.013\text{g}}{23.9\text{L}}$$

$$d = 1.84 \text{ g/L}$$

Guided Solution

[Download Guided Solution as a pdf](#)

	Guided Solution	Hint
tion	This is a calculation type problem where you are required to calculate the density of a given compound using one of the two possible methods.	See LibreText 2.4.1 (section 2.4.1 Density of a Gas)
tion	Question: What is the density of laughing gas, dinitrogen monoxide, N ₂ O, at a temperature of 325 K and a pressure of 113.0 kPa?	We are given the temperature in Kelvin , the pressure in kilopascals and the chemical formula of Dinitrogen Monoxide. From this information, we are trying to calculate the <u>density of the dinitrogen monoxide</u> .
tion	You will need to convert from kilopascals to atmospheres. The conversion value is: 101.325kPa = 1atm.	You will need to convert from kilopascals to atmospheres. The conversion factor is: 101.325kPa = 1atm.
	Recall how to determine density from mass and volume.	$d = \frac{m}{V}$
tion	How to manipulate the equation to derive a new one: 1. Recall the density equation. 2. Recall n is amount in grams per mole which is mass/molar mass. 3. Substitute ($\frac{\text{mass}}{\text{molar mass}}$) in the ideal gas equation in place of n. 4. Manipulate the ideal gas equation (move the variables around) so that it is equal to density.	We know the equation for density is $d = \frac{m}{V}$ We know the ideal gas equation is $PV = nRT$ We know that $n = \frac{\text{mass}}{\text{molar mass}} = \frac{m}{M}$ Therefore, we can substitute $\frac{m}{M}$ in the ideal gas equation in place of n. $PV = \frac{m}{M}RT$ Now, we can manipulate the equation to be equal to $\frac{m}{V}$ which is equal to density (d). $d = \frac{m}{V} = \frac{PM}{RT}$
tion	OR use the equations you already know: Using the ideal gas equation, plug in the information your given to solve for volume. THEN using the density equation, plug in your calculated volume. What about the mass?	Recall that you can find the mass by finding the molar mass of the compound. Look at the chemical formula.

tion

Guided Solution

Hint

Complete Solution:

$$113.0 \text{ kPa} = \frac{1 \text{ atm}}{101.325 \text{ kPa}} = 1.115 \text{ atm}$$

Approach a, using a derived equation:

$$\text{We know the equation for density is } d = \frac{\text{mass}}{\text{volume}} = \frac{m}{V}$$

$$\text{We know the ideal gas equation is } PV = nRT$$

Now, we can manipulate the equation to be equal to $\left(\frac{\text{mass}}{\text{volume}}\right)$ which is equal to density (d).

$$d = \frac{m}{V} = \frac{PM}{RT}$$

Plugging into the equation:

$$d = \frac{PM}{RT}$$

You need the molar mass of N_2O :

$$M = 2(14.007) + 15.999 = 44.013 \text{ g/mol}$$

$$d = \frac{(1.115 \text{ atm})(44.013 \text{ g/mol})}{(0.08206 \text{ atm} \cdot \text{L/mol} \cdot \text{k})(325 \text{ k})}$$

$$d = 1.84 \text{ g/L}$$

Approach b, using a two steps:

1. Calculate the volume.

Plug known information into the ideal gas equation; rearranged to solve for volume.

$$PV = nRT$$

$$V = \frac{nRT}{P}$$

$$V = \frac{(1.000 \text{ mol})(0.08206 \text{ atm} \cdot \text{L/mol} \cdot \text{k})(325 \text{ k})}{1.115 \text{ atm}}$$

$$V = 23.9 \text{ L}$$

2. Calculate the density.

$$d = \frac{m}{V}$$

Using the volume you calculated in the previous step, and the molar mass of N_2O plug into the density equation:

$$d = \frac{44.013 \text{ g}}{23.9 \text{ L}}$$

$$d = 1.84 \text{ g/L}$$

Answer 1.84 g/L

How to find M (molar mass) of N_2O :

Look up element masses on periodic table, then add up making sure you use molecule stoichiometry.

$$N = 14.007 \text{ g/mol} \times 2$$

$$O = 15.999 \text{ g/mol}$$

$$\text{Molar mass} = 2(14.007) + 15.999 = 44.013 \text{ g/mol}$$

tion

Check your work!

Make sure your units cancel when you plug them in to your equation, and that your final result is in the appropriate units for density.

Why does this answer make chemical sense?

Density is the amount of matter that is packed into a given volume (mass divided by volume). It is a physical property that varies between materials. Gas density is the amount of matter in each volume at a specific temperature and pressure, therefore it makes sense that we can calculate the density of dinitrogen monoxide gas at these specific conditions. It does not matter which stage you include the conditions in your calculation if they are used to achieve your final solution.

Using a dimensional analysis approach and writing units beside values in equations lets you see if the units cancel to give you a result in appropriate units. Our gas density should have units of g/L.

(question from page titled 9.E Gases (Exercises), shared under a CC BY 4.0 license platform [https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_1e_\(OpenSTAX\)/09%3AGases/9.E%3AGases_\(Exercises\)](https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_1e_(OpenSTAX)/09%3AGases/9.E%3AGases_(Exercises))), original source <https://openstax.org/books/chemistry-2e/pages/9-exercises>, access for free at <https://openstax.org/books/chemistry-2e/pages/1-introduction>)

2.2: PASS Ideal Gases - Calculate the density of an ideal gas (2.E.28) is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

2.3: PASS Ideal Gases - partial pressure, gas over water calculation (2.E.36)

This work by Jensen, Brewer, Blackstock is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



? Exercise 2.3.2.36

A sample of carbon monoxide was collected over water at a total pressure of 756 torr and a temperature of 18 °C. What Is the pressure of the carbon monoxide? The vapour pressure of H₂O at 18°C is 15.5 torr.

Answer

741 torr

Strategy Map

	Step	Hint
	1. Identify what gas the question is asking you to find the pressure for.	You have a gas mixture. See LibreText 2.4.3 The pressure of a mixture of gases.
ify ...	2. Identify the total pressure and the vapour pressure.	Recall vapour pressure is the partial pressure of water (see LibreText section 2.4.3.1)
ify ...	3. Use the sum of the partial pressures equation.	

Solution

$$\begin{aligned}
 P_{\text{tot}} &= P_{\text{gas}} + P_{\text{H}_2\text{O}} \\
 756\text{torr} &= P_{\text{gas}} + 15.5\text{torr} \\
 756\text{torr} - 15.5\text{torr} &= p_{\text{gas}} \\
 P_{\text{gas}} &= \mathbf{740.5 \text{ torr}} \\
 &\text{or} \\
 P_{\text{gas}} &= \mathbf{741 \text{ torr}}
 \end{aligned}$$

Guided Solution

[Download Guided Solution as a pdf](#)

	Guided Solution	Hint
tion	This is a theory type problem where you must use your knowledge of partial pressures to find the pressure of carbon monoxide gas when it is collected over water.	See LibreText 2.4.3 The pressure of a mixture of gases.
tion	What does it mean by collected over water?	Recall that when a gas is collected over water, you have a mixture of the collected gas and water vapour. The vapour pressure (the partial pressure of water) must be subtracted from the total pressure.
tion	Recall how to calculate the SUM of all partial pressures.	$P_{\text{tot}} = P_{\text{gas}} + P_{\text{H}_2\text{O}}$ (see LibreText section 2.4.3.1)

	Guided Solution	Hint
tion	Complete Solution: Recall that the sum of your partial pressures is equal to your total pressure. $P_{\text{tot}} = P_{\text{gas}} + P_{\text{H}_2\text{O}}$ Plug in the values provided in the question. $756 \text{ torr} - 15.5 \text{ torr} = p_{\text{gas}}$ Isolate and solve for your desired variable. $756 \text{ torr} - 15.5 \text{ torr} = p_{\text{gas}}$ $P_{\text{gas}} = 740.5 \text{ torr}$ or $P_{\text{gas}} = 741 \text{ torr}$ Answer 741 torr	The answer should have no decimal places since the one of the values in the question does not have any digits after the decimal.
tion	Check your work! Your partial pressure of carbon dioxide should be less than your total pressure. Why does this answer make chemical sense? When you are collecting a gas over water, that water is constantly changing between phases from liquid to solid. When you collected your desired gas, you are collecting both the desired gas as well as some water molecules in its gaseous state. For this reason, you must subtract the partial pressure of water (the vapour pressure) from your total calculated pressure. This will give you the pressure of only your desired gas.	If your CO₂ pressure is higher than the total pressure then you have likely added instead of subtracting the water vapour pressure.

(question source from page titled 9.11: Exercises, shared under a CC BY 4.0 license platform [https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_2e_\(OpenSTAX\)/09%3AGases/9.11%3AExercises](https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_2e_(OpenSTAX)/09%3AGases/9.11%3AExercises), shared under a CC BY 4.0 license, authored, remixed, and/or curated by OpenStax, original source <https://openstax.org/books/chemistry-2e/pages/9-exercises>), Access for free at <https://openstax.org/books/chemistry/pages/1-introduction>)

2.3: PASS Ideal Gases - partial pressure, gas over water calculation (2.E.36) is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

2.4: PASS Ideal Gases- use the Kinetic Molecular Theory to explain impact on a gas as V and T increased (2.E.45)

This work by Jensen, Brewer, Blackstock is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



? Exercise 2.E.45

A 1 L sample of CO initially at 0°C and 1atm is heated to 546 K, and its volume is increased to 2 L.

1. What effect do these changes have on the number of collisions of the molecules of the gas per unit area of the container wall?
2. What is the effect on the average kinetic energy of the molecules?
3. What is the effect on the root mean square speed of the molecules?

Answer

1. The number of collisions per unit area of the container wall is constant.
2. The average kinetic energy doubles.
3. The root mean square speed increases $\sqrt{2}$ times its initial value; U_{rms} is proportional to KE_{avg} .

Strategy Map

	Step	Hint
	1. Recall the relationships between temperature, pressure, and volume.	As Temperature increases, Volume increases. As Temperature increases, Pressure increases. As Volume increases, Pressure decreases.
I th...	2. Recall the relationship between temperature and kinetic energy.	As Temperature increases, the kinetic energy of gas molecules increases. See LibreText section 2.5 The Kinetic-Molecular Theory
I th...	3. Recall the relationship to determine average kinetics energy and what change would cause it to increase or decrease.	$KE_{avg} = \frac{3}{2} RT$
I th...	4. Recall what root mean square speed is and what causes it to increase or decrease.	$U_{rms} = \sqrt{\frac{3RT}{M(x)}}$
I th...	5. Look at the question information and determine which parameters change and how they are related.	$V_1 = 1 \text{ L}; V_2 = 2 \text{ L}$ $T_1 = 0^\circ\text{C} = 273 \text{ K}; T_2 = 546 \text{ K}$ n is constant What happens to P ?

Solution

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$T_2 = 2 T_1$$

$$V_2 = 2 V_1$$

Substitute in to the equation for changing conditions:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 2 V_1}{2 T_1}$$

Cancel terms that are the same

$$\frac{P_1 V_1}{T_1} = \frac{P_2 2 V_1}{2 T_1}$$

Left with $P_1 = P_2$, no change in pressure.

answer 1. Since pressure does not change, the number of collisions per unit area of the container wall is constant.

Looking at the impact of increased temperature on kinetic energy, as T doubles kinetic energy doubles.

$$KE_{\text{avg}} = \frac{3}{2} RT$$

answer 2. Since T_2 is 2 times T_1 , the average kinetic energy doubles.

Looking at the impact of increased temperature on root mean square speed, as T doubles root mean square speed increase by $\sqrt{2}$ times.

$$U_{\text{rms}} = \sqrt{\frac{3RT}{M(x)}}$$

$$U_{\text{rms}} = \sqrt{\frac{3RT_1}{M(x)}} \text{ or } U_{\text{rms}} = \sqrt{\frac{3R(2T_1)}{M(x)}}$$

answer 3. The root mean square speed increases since the temperature has increased, it increases $\sqrt{2}$ times its initial value; U_{rms} is proportional to KE_{avg} .

Guided Solution

Download Guided Solution as a pdf

	Guided Solution	Hint
tion	This is a theory type problem that tests your knowledge on the kinetic molecular theory. The temperature and volume of a gas is increased, you must use your knowledge on the relationships of this chapter to predict the kinetics of the gas.	See LibreText section 2.5 The Kinetic-Molecular Theory
tion	A 1 L sample of CO initially at 0°C and 1atm is heated to 546 K, and its volume is increased to 2 L. 1. What effect do these changes have on the number of collisions of the molecules of the gas per unit area of the container wall? 2. What is the effect on the average kinetic energy of the molecules? 3. What is the effect on the root mean square speed of the molecules?	As the temperature of the gas was increased from 273 K to 546 K, the volume of the gas increased from 1L to 2L.
tion	Recall your relationships	- As temperature increases, pressure increases. - As temperature increases, volume increases. - As temperature increases, kinetic energy of gas molecules increases. - As volume increases, pressure decreases.
tion	Recall the equations that demonstrate these relationships	$PV = nRT$ $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ $KE_{\text{avg}} = \frac{3}{2} RT$
tion	What is “root mean square speed” and what equation is used to calculate it?	The root mean square speed of a particle, U_{rms}, is defined as the square root of the average of the squares of the velocities with n = the number of particles. $U_{\text{rms}} = \sqrt{\frac{3RT}{M(x)}}$

tion

Guided Solution

Hint

Complete Solution:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$T_2 = 2 T_1$$

$$V_2 = 2 V_1$$

Substitute in to the equation for changing conditions:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 2 V_1}{2 T_1}$$

Cancel terms that are the same:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 2 V_1}{2 T_1}$$

Left with $P_1 = P_2$, no change in pressure.

The pressure is the number of collisions per unit of area, if the pressure is constant, so are the number of collisions.

answer 1. Since pressure does not change, the number of collisions per unit area of the container wall is constant.

$$KE_{avg} = \frac{3}{2} RT$$

As the temperature increases, so does the average kinetic energy.

answer 2. Since T_2 is 2 times T_1 , the average kinetic energy doubles.

$$U_{rms} = \sqrt{\frac{3RT}{M(x)}}$$

$$U_{rms} = \sqrt{\frac{3RT_1}{M(x)}} \text{ or } U_{rms} = \sqrt{\frac{3R(2T_1)}{M(x)}}$$

If the temperature increases, and the number of particles stays constant, the speed will increase.

answer 3. The root mean square speed increases since the temperature has increased, it increases $\sqrt{2}$ times its initial value; U_{rms} is proportional to KE_{avg} .

tion

Check your work!

Make sure you look at the impact of increasing temperature and increasing volume on the parameters you are asked about in the question.

To evaluate impact on collisions you must first think about impact on pressure using the appropriate form of the general gas law for changing conditions.

Since T increases we expect KE_{avg} and U_{rms} to increase.

Why does this answer make chemical sense?

If one factor causes an increase in pressure while the other causes a decrease in the pressure, they will counteract one another. In this case each increased by a factor of 2 which cancels. So the pressure will stay constant. If factors cause the speed to increase, and nothing impacts the speed to decrease, the particles will move faster. This can be seen by the equations below.

$$KE_{avg} = \frac{3}{2} RT$$

This shows that kinetic energy and temperature are proportional.

$$U_{rms} = \sqrt{\frac{3RT}{M(x)}}$$

This shows that the root mean square speed is dependent on temperature.

See [LibreText section 2.5 The Kinetic-Molecular Theory](https://chem.libretexts.org/@page/467006)

(question source from page titled 9.E:Gases (Exercises)
[https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_\(OpenSTAX\)/09%3A_Gases/9.E%3A_Gases_\(Exercises\)](https://chem.libretexts.org/Bookshelves/General_Chemistry/Chemistry_(OpenSTAX)/09%3A_Gases/9.E%3A_Gases_(Exercises)),
shared under a CC BY 4.0 license, authored, remixed, and/or curated by OpenStax, original source
<https://openstax.org/books/chemistry/pages/9-exercises>, Access for free at <https://openstax.org/books/chemistry/pages/1-introduction>)

2.4: PASS Ideal Gases- use the Kinetic Molecular Theory to explain impact on a gas as V and T increased (2.E.45) is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

CHAPTER OVERVIEW

3: Thermochemistry

Topic hierarchy

3.1: PASS Thermochemistry: calculate q for cooling a compound (3.E.14)

3.2: PASS Thermochemistry- Interpreting thermochemical equations and sketching enthalpy diagrams (3.E.10a)

3: Thermochemistry is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

3.1: PASS Thermochemistry: calculate q for cooling a compound (3.E.14)

This work by Jensen, Brewer, Blackstock is licensed under [CC BY-NC 4.0](#)



? Exercise 3.1.3E.14

Question

Ethylene glycol, used as a coolant in automotive engines, has a specific heat capacity of $2.42 \text{ J g}^{-1} \text{ K}^{-1}$. Calculate q for the system when $3.65 \times 10^3 \text{ g}$ of ethylene glycol is cooled from 115.0°C to 85.0°C .

Answer

$$q = -265 \text{ kJ}$$

Refer to [LibreText 3.5, Calorimetry](#).

Strategy Map

Step	Hint
1. Identify what information you are given in the question.	specific heat capacity initial and final temperature mass of ethylene glycol
2. Identify what the question is asking you to solve for.	q when ethylene glycol is cooled
3. Recall an equation that connects the provided information and what you are looking for.	You are given the specific heat capacity. It has the units J/gK . Use these units to help recall your calorimetry equation.

Solution

$$\begin{aligned}
 q &= c_p m \Delta T \\
 q &= \left(2.42 \frac{\text{J}}{\text{gK}} \right) (3.65 \times 10^3 \text{ g}) (85.0^\circ\text{C} - 115.0^\circ\text{C}) \\
 q &= \left(2.42 \frac{\text{J}}{\text{gK}} \right) (3.65 \times 10^3 \text{ g}) (-30.0 \text{ K}) \\
 q &= -264990 \text{ J} \\
 q &= -264.990 \text{ kJ} \\
 q &= -265 \text{ kJ}
 \end{aligned}$$

Guided Solution

[Download Guided Solution as a pdf](#)

Guided Solution

This is a calculation problem where your knowledge of calorimetry is tested by giving you experimental data to solve for the heat (q) lost by ethylene glycol when it cools.

Hint

Refer to: [LibreText 3.5, Calorimetry](#)

	Guided Solution	Hint
tion	Ethylene glycol, used as a coolant in automotive engines, has a specific heat capacity of $2.42 \text{ J g}^{-1} \text{ K}^{-1}$. Calculate q when $3.65 \times 10^3 \text{ g}$ of ethylene glycol is cooled from 115.0°C to 85.0°C .	Provided Information: Specific heat capacity (c_p) = 2.42 J/gk Mass $m = 3.65 \times 10^3 \text{ g}$ Temperature Change (ΔT) = Final T – Initial T $\Delta T = T_f - T_i = 85.0^\circ\text{C} - 115.0^\circ\text{C} = 30.0^\circ\text{C} = 30.0 \text{ K}$ Looking for: q, heat lost by ethylene glycol when it cools
tion	If the Ethylene glycol is cooled down, what can you expect about your final answer?	We are looking for q , the heat gained or lost by the system. If the system is cooling down, it is losing energy in the form of heat. This means the final answer for q will be negative.
tion	Recall the different equations for calorimetry. Since we are using specific heat capacity, we will use the equation with c_p .	You are given the specific heat capacity. It has the units J/gk use these units to help recall your calorimetry equation. Specific heat capacity (c_p) = 2.42 J/g k $\text{J} = \text{Joules} = \text{unit of energy} = q$ $\text{g} = \text{grams} = \text{unit of mass} = m$ $\text{K} = \text{Kelvin} = \text{unit of temperature} = \Delta T$ $c_p = \frac{q}{m\Delta T}$ Then rearrange for q, $q = c_p m \Delta T$
tion	Complete Solution	
tion	Recall the calorimetry equation that uses specific heat capacity: $\mathrm{q} = \mathrm{c}_p \mathrm{m} \Delta \mathrm{T}$ Use the provided information to plug into the equation. $(c_p) = 2.42 \text{ J/gk}$ $(m) = 3.65 \times 10^3 \text{ g}$ $(\Delta T) = T_f - T_i = 85.0^\circ\text{C} - 115.0^\circ\text{C} = 30.0^\circ\text{C}$ $q = \left(2.42 \frac{\text{J}}{\text{gk}} \right) (3.65 \times 10^3 \text{ g}) (85.0^\circ\text{C} - 115.0^\circ\text{C})$ $q = \left(2.42 \frac{\text{J}}{\text{gk}} \right) (3.65 \times 10^3 \text{ g}) (-30.0 \text{ K})$ Multiply out and solve for q. $q = -264990 \text{ J}$ Recall that the significant figures in your final answer should match the information given in the question and that q should be in kilojoules. $q = -264.990 \text{ kJ}$ $q = -265 \text{ kJ}$	

tion

Guided Solution

Hint

Check your work!

Since ethylene glycol (the system), is cooling down, it is losing energy in the form of heat. This means ΔT is negative, and we expect the final answer for q will be negative.

Why does this answer make chemical sense?

Calorimetry is the process of calculating the energy change (or enthalpy change) during a chemical reaction. This is done by measuring the amount of heat lost or gained by the system.

With the data provided in the question we know that in this situation the system was losing heat and therefore its energy change was negative. From this we know the process was exothermic and released -265 kJ into its surroundings.

Make sure you calculate ΔT correctly.

$$\Delta T = T_f - T_i$$

(question source from page titled 3.E: Thermochemistry (Exercises), [https://chem.libretexts.org/Courses/Thompson_Rivers_University/TRU%3A_Fundamentals_and_Principles_of_Chemistry_\(CHEM_1510_and_CHEM_1520\)/03%3A_Thermochemistry/3.E%3A_Thermochemistry_\(Exercises\)](https://chem.libretexts.org/Courses/Thompson_Rivers_University/TRU%3A_Fundamentals_and_Principles_of_Chemistry_(CHEM_1510_and_CHEM_1520)/03%3A_Thermochemistry/3.E%3A_Thermochemistry_(Exercises))), shared under a CC BY-NC-SA 3.0 license, authored, remixed, and/or curated by LibreTexts.

3.1: PASS Thermochemistry: calculate q for cooling a compound (3.E.14) is shared under a not declared license and was authored, remixed, and/or curated by LibreTexts.

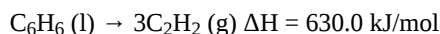
3.2: PASS Thermochemistry- Interpreting thermochemical equations and sketching enthalpy diagrams (3.E.10a)

This work by Jensen, Brewer, Blackstock is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



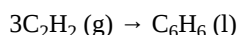
Question

Given the following thermochemical equation,

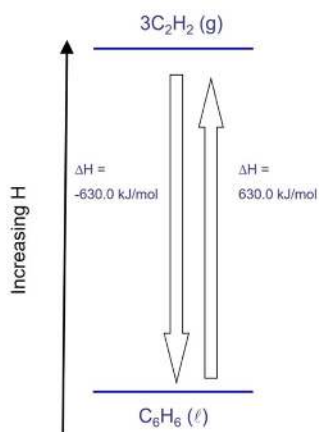


Write the complete thermochemical equation for the reverse reaction. Sketch an enthalpy diagram showing the relative positions of all species and include arrows indicating both of the above reactions with appropriate values for ΔH reaction.

Answer



$$\Delta\text{H} = -630.0 \text{ kJ/mol}$$

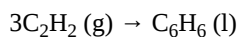


See LibreText section 3.4 (Enthalpy of Reaction)

Strategy Map

	Step	Hint
	1. Identify the reactants and products for the forward reaction.	
t...	2. Swap them to give the reverse reaction.	Recall reversing a reaction changes the sign of ΔH (see LibreText section 3.4)
t...	3. Manipulate the enthalpy value from the forward reaction to give the value for the reverse reaction.	The amount of energy will be the same regardless of the reaction's direction. However, in one direction energy is absorbed and in the other energy is released.
t...	4. Create the enthalpy diagram showing the reaction in both directions.	

Solution



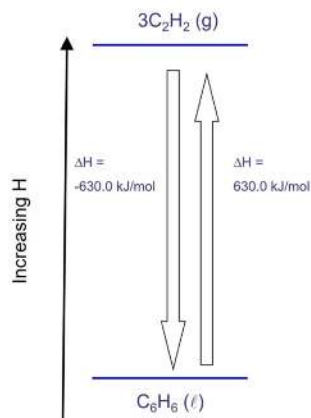
The reactant and product from the forward reaction switch places.

$$\Delta H = -630.0 \text{ kJ/mol}$$

The sign of the enthalpy value for the forward reaction is flipped.

The thermochemical equation for the reverse reaction is: $3\text{C}_2\text{H}_2(\text{g}) \rightarrow \text{C}_6\text{H}_6(\text{l}) \Delta H = -630.0 \text{ kJ/mol}$

The enthalpy diagram shows the relative positions of all species and includes arrows indicating both of the above reactions with



appropriate values for ΔH reaction.

The up arrow represents the forward endothermic reaction, and the down arrow represents the reverse exothermic reaction.

Guided Solution

[Download Guided Solution as a pdf](#)

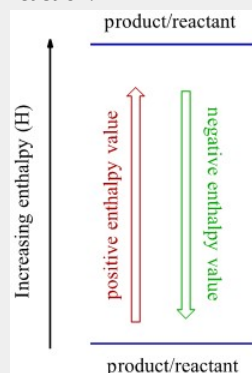
	Guided Solution	Hint
m	Question Given the following thermochemical equation, $\text{C}_6\text{H}_6(\text{l}) \rightarrow 3\text{C}_2\text{H}_2(\text{g}) \Delta H = 630.0 \text{ kJ/mol}$ Write the complete thermochemical equation for the reverse reaction. Sketch an enthalpy diagram showing the relative positions of all species and include arrows indicating both of the above reactions with appropriate values for ΔH reaction.	
m	This is a theory type problem where your knowledge on enthalpy is tested. You must find the reverse reaction given its forward reaction, find its enthalpy and sketch the diagram that represents the reaction in both directions.	See LibreText 3.4: Enthalpy of Reaction
m	You are given information about the forward reaction; how can it be manipulated to give the reverse reaction?	Recall that if a reaction is reversed, the reactants and products switch sides and the opposite reaction is taking place.
m	What about the enthalpy?	The magnitude of energy will be the same regardless of the reactions direction. However, in one direction energy is absorbed and in the other energy is released.
m	If the enthalpy change for the forward reaction is positive, what do you expect the enthalpy change for the reverse reaction to be?	If the enthalpy for the forward reaction is positive you can expect the enthalpy for the reverse reaction to be negative .

Guided Solution

Hint

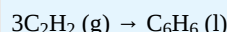
- Recall how to sketch an enthalpy diagram...
1. On the lefthand side draw an arrow and label its direction with the corresponding enthalpy change (either increasing or decreasing).
 2. Draw your reactants and an arrow leading to the products.

If the reaction is endothermic (it has a positive enthalpy) the products will be higher than the reactants and the reaction arrow will point upwards. The opposite is true for an exothermic reaction.



- Complete Solution:**
Given: $\text{C}_6\text{H}_6(\text{l}) \rightarrow 3\text{C}_2\text{H}_2(\text{g}) \Delta H = 630.0 \text{ kJ/mol}$

Reverse the reaction



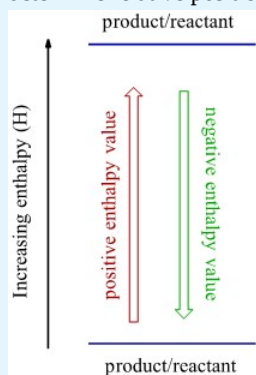
The reactant and product from the forward reaction switch places. In the forward reaction three C_2H_2 molecules break to form one C_6H_6 molecule. In the reverse reaction one C_6H_6 molecule is broken to form three C_2H_2 molecules.

Change the sign of ΔH

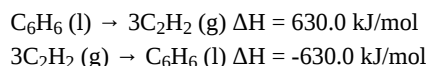
$$\Delta H = 630.0 \text{ kJ/mol}$$

The sign of the enthalpy value for the forward reaction is flipped to give the enthalpy value of the reverse reaction. This will always be the case since one way is endothermic while the other is exothermic.

Now use the enthalpy diagram template to position the reactant and product, paying attention to the sign of ΔH in each case to determine relative positions.



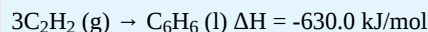
The forward and reverse reactions are placed on the enthalpy diagram. The upward arrow represents the forward reaction, and the downward arrow represents the reverse reaction. The forward reaction is endothermic meaning it requires an input of energy. The reverse reaction is an exothermic reaction meaning energy is released from the reaction.



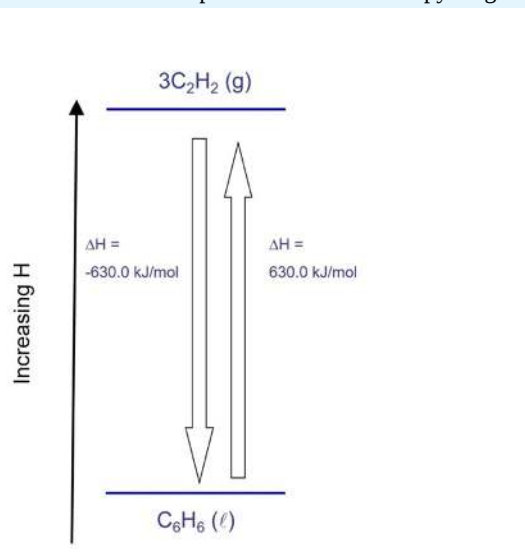
Guided Solution

Hint

Answer The complete thermochemical equation for the reverse reaction is therefore:



Both reactions are represented on the enthalpy diagram:



Check your work!

A complete thermochemical equation has reactant, product and value of ΔH .

The reaction given in the question was endothermic, so it is represented by the up arrow on the enthalpy diagram. The reverse reaction is exothermic, so it is represented as the up arrow.

Why does this answer make chemical sense?

For a chemical reaction, the enthalpy of reaction (ΔH) is the difference in enthalpy between products and reactants. Reversing a chemical reaction reverses the sign of the enthalpy. This makes sense because for a reaction to occur in one direction it will be either exothermic or endothermic when the bonds break and form. Therefore, if the same reaction happens but the opposite bonds are breaking and forming the enthalpy will be in the opposite direction. This means that if the forward reaction is endothermic the reverse reaction will be exothermic and vice versa.

Make sure that your 'down' arrow represents the exothermic reaction and your 'up' arrow represents the endothermic reaction.

(Original question authored by Sharon Brewer shared under a [CC BY-NC-SA 4.0 license](https://creativecommons.org/licenses/by-nc-sa/4.0/))

3.2: PASS Thermochemistry- Interpreting thermochemical equations and sketching enthalpy diagrams (3.E.10a) is shared under a [not declared license](#) and was authored, remixed, and/or curated by LibreTexts.

CHAPTER OVERVIEW

4: Kinetics

4: Kinetics is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

Detailed Licensing

Overview

Title: PASS Chemistry Book CHEM 1510/1520

Webpages: 35

Applicable Restrictions: Noncommercial

All licenses found:

- [Undeclared](#): 57.1% (20 pages)
- [CC BY-NC-SA 4.0](#): 42.9% (15 pages)

By Page

- [PASS Chemistry Book CHEM 1510/1520 - CC BY-NC-SA 4.0](#)
 - [Front Matter - Undeclared](#)
 - [TitlePage - Undeclared](#)
 - [InfoPage - Undeclared](#)
 - [Table of Contents - Undeclared](#)
 - [Licensing - Undeclared](#)
 - [1: Introduction - CC BY-NC-SA 4.0](#)
 - [1.1: About PASS and the Creators - Undeclared](#)
 - [1.2: How to use PASS Chem - CC BY-NC-SA 4.0](#)
 - [2: Gases - CC BY-NC-SA 4.0](#)
 - [2.1: PASS Ideal Gases- General gas law calculation, changing temperature \(2.E.12\) - CC BY-NC-SA 4.0](#)
 - [2.2: PASS Ideal Gases - Calculate the density of an ideal gas \(2.E.28\) - CC BY-NC-SA 4.0](#)
 - [2.3: PASS Ideal Gases - partial pressure, gas over water calculation \(2.E.36\) - Undeclared](#)
 - [2.4: PASS Ideal Gases- use the Kinetic Molecular Theory to explain impact on a gas as V and T increased \(2.E.45\) - CC BY-NC-SA 4.0](#)
 - [3: Thermochemistry - CC BY-NC-SA 4.0](#)
 - [3.1: PASS Thermochemistry: calculate q for cooling a compound \(3.E.14\) - Undeclared](#)
 - [3.2: PASS Thermochemistry- Interpreting thermochemical equations and sketching enthalpy diagrams \(3.E.10a\) - Undeclared](#)
 - [4: Kinetics - CC BY-NC-SA 4.0](#)
 - [4.1: PASS Kinetics - Writing Rate Expressions - CC BY-NC-SA 4.0](#)
 - [5: Chemical Equilibrium - CC BY-NC-SA 4.0](#)
 - [5.1: Template draft - Undeclared](#)
 - [6: Acid-Base Equilibrium - CC BY-NC-SA 4.0](#)
 - [6.1: Template draft - Undeclared](#)
 - [7: Buffers, Titrations and Solubility Equilibria - CC BY-NC-SA 4.0](#)
 - [7.1: Template draft - Undeclared](#)
 - [8: Entropy - CC BY-NC-SA 4.0](#)
 - [8.1: Template draft - Undeclared](#)
 - [9: Electrochemistry - CC BY-NC-SA 4.0](#)
 - [9.1: Page Title - Undeclared](#)
 - [9.2: Template draft - Undeclared](#)
 - [Back Matter - Undeclared](#)
 - [Index - Undeclared](#)
 - [Glossary - Undeclared](#)
 - [Detailed Licensing - Undeclared](#)
 - [Detailed Licensing - Undeclared](#)

Detailed Licensing

Overview

Title: PASS Chemistry Book CHEM 1510/1520

Webpages: 35

Applicable Restrictions: Noncommercial

All licenses found:

- [Undeclared](#): 57.1% (20 pages)
- [CC BY-NC-SA 4.0](#): 42.9% (15 pages)

By Page

- [PASS Chemistry Book CHEM 1510/1520 - CC BY-NC-SA 4.0](#)
 - [Front Matter - Undeclared](#)
 - [TitlePage - Undeclared](#)
 - [InfoPage - Undeclared](#)
 - [Table of Contents - Undeclared](#)
 - [Licensing - Undeclared](#)
 - [1: Introduction - CC BY-NC-SA 4.0](#)
 - [1.1: About PASS and the Creators - Undeclared](#)
 - [1.2: How to use PASS Chem - CC BY-NC-SA 4.0](#)
 - [2: Gases - CC BY-NC-SA 4.0](#)
 - [2.1: PASS Ideal Gases- General gas law calculation, changing temperature \(2.E.12\) - CC BY-NC-SA 4.0](#)
 - [2.2: PASS Ideal Gases - Calculate the density of an ideal gas \(2.E.28\) - CC BY-NC-SA 4.0](#)
 - [2.3: PASS Ideal Gases - partial pressure, gas over water calculation \(2.E.36\) - Undeclared](#)
 - [2.4: PASS Ideal Gases- use the Kinetic Molecular Theory to explain impact on a gas as V and T increased \(2.E.45\) - CC BY-NC-SA 4.0](#)
 - [3: Thermochemistry - CC BY-NC-SA 4.0](#)
 - [3.1: PASS Thermochemistry: calculate q for cooling a compound \(3.E.14\) - Undeclared](#)
 - [3.2: PASS Thermochemistry- Interpreting thermochemical equations and sketching enthalpy diagrams \(3.E.10a\) - Undeclared](#)
 - [4: Kinetics - CC BY-NC-SA 4.0](#)
 - [4.1: PASS Kinetics - Writing Rate Expressions - CC BY-NC-SA 4.0](#)
 - [5: Chemical Equilibrium - CC BY-NC-SA 4.0](#)
 - [5.1: Template draft - Undeclared](#)
 - [6: Acid-Base Equilibrium - CC BY-NC-SA 4.0](#)
 - [6.1: Template draft - Undeclared](#)
 - [7: Buffers, Titrations and Solubility Equilibria - CC BY-NC-SA 4.0](#)
 - [7.1: Template draft - Undeclared](#)
 - [8: Entropy - CC BY-NC-SA 4.0](#)
 - [8.1: Template draft - Undeclared](#)
 - [9: Electrochemistry - CC BY-NC-SA 4.0](#)
 - [9.1: Page Title - Undeclared](#)
 - [9.2: Template draft - Undeclared](#)
 - [Back Matter - Undeclared](#)
 - [Index - Undeclared](#)
 - [Glossary - Undeclared](#)
 - [Detailed Licensing - Undeclared](#)
 - [Detailed Licensing - Undeclared](#)